

GONZALO LOAYZA

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SUMMARY

Senior Computer Science student at BYU with a Machine Learning emphasis and applied experience in data engineering, industrial analytics, time-series, simulation, and decision-support workflows. Skilled in Python, SQL, feature engineering, model evaluation, and translating real-world operational data into practical technical insights.

EDUCATION

Brigham Young University (BYU), Provo, UT

B.S. Computer Science, Machine Learning Emphasis | Expected Dec 2026 | BYU Honors Program

Coursework: Deep Learning, Data Science Capstone, Machine Learning, Algorithms, Data Structures, Computer Systems, Probability & Statistics | Honors Thesis: Operational Mode Discovery and Business Value Analysis in Industrial Time-Series Data

TECHNICAL SKILLS

Languages: Python, SQL, C++, JavaScript | ML & Data: time-series analysis, feature engineering, classification, unsupervised learning, PCA, model evaluation, EDA, predictive analytics | Tools: scikit-learn, Pandas, NumPy, Jupyter, Git, Linux, SHAP, Power BI | Applied Analytics: KPI analysis, bottleneck analysis, operational mode discovery, value-driver identification, decision support

EXPERIENCE

Data Analytics Intern

Hatch Digital, Peru | Jun 2026 - Aug 2026

- Supported digital and analytics initiatives for mining and industrial operations, including tailings-data automation, operational KPIs, measurement concepts, and decision-support workflows.
- Contributed to problem framing, data preparation, dashboard-oriented analysis, and identification of opportunities for process improvement and operational value creation.

Maintenance Data Analyst Intern

Compañía de Minas Buenaventura, Peru | Jun 2025 - Aug 2025

- Processed and cleaned 50k+ noisy sensor time-series from crushing and grinding equipment; engineered 15+ reliability features including temperature deltas, load ratios, and transient-spike indicators.
- Analyzed equipment wear, ore variability, and process signals for condition-monitoring workflows; structured 100+ assets under ISO 14224/17359 to improve traceability and analytical usability.

Data Analytics Intern

Hatch Ltd, Urban Solutions Sector, Vermont, USA | May 2024 - Aug 2024

- Analyzed 200k+ rail-simulation records to evaluate delays, network performance, and operational bottlenecks; built Python and C++ automation that reduced output-processing time by about 70%.
- Integrated rider-survey information with onboard sensor data and supported scenario planning through statistical and optimization-based analysis.

SELECTED PROJECTS

Honors Thesis - Operational Mode Discovery & Business Value Analysis

Python | Industrial Time-Series | PCA | Clustering | BYU Honors, 2025-2026

- Integrated minute-level process data with daily KPI context, then applied preprocessing, scaling, PCA, and clustering to identify recurrent operating modes in a real industrial processing environment.
- Connected operating modes, control performance, instrumentation gaps, and ML opportunities to performance stability and business-value drivers.

Wildfire Prediction System

Python | XGBoost | SHAP | APIs | BYU, 2026

- Built a wildfire-risk prototype using seven-day weather time-series and reported fire data; compared XGBoost, Random Forest, and Naive Bayes on imbalanced event data.
- Applied SHAP interpretability and designed a workflow combining weather APIs, satellite-imagery retrieval, and dashboard-style risk visualization.

AWARDS, LEADERSHIP & LANGUAGES

Awards: Donald Goodyear Doll Sr. Scholarship; Dr. Gerald Hatch Scholarship; BYU Honors Program | Leadership: Emergency Response & Rescue Program, Peruvian Army | Languages: English (Fluent), Spanish (Native), French (Intermediate)